

PONMUDI M R

Electronics & Communication Engineering

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CAREER OBJECTIVE

Motivated ECE undergraduate seeking internships and research opportunities in embedded systems, RF communication, and IoT. Eager to apply hands-on project experience and technical problem-solving skills to contribute to innovative engineering solutions at technology-driven organizations.

EDUCATION

B.E. in Electronics & Communication Engineering | *Bannari Amman Institute of Technology* 2025 – 2029
Sathyamangalam, Tamil Nadu

Higher Secondary (Class XII) | *CBSE Board* Completed
Tamil Nadu, India

TECHNICAL PROJECTS

RF-Based Disaster Communication & Tracking System | *LoRa + ESP32* 2025

- ▶ Engineered an off-grid RF communication system enabling data transmission in network-denied environments such as disaster zones and remote areas
- ▶ Implemented LoRa (Long Range) protocol with ESP32 microcontroller for low-power, long-distance wireless communication up to 15+ km range
- ▶ Integrated GPS tracking module for real-time location monitoring and emergency beacon functionality
- ▶ Awarded 1st Prize at NHIDE 2026 National Innovation Hackathon – recognized for technical innovation and practical implementation

Smart Fabric Length Measurement System | *Industrial IoT Solution* In Progress

- ▶ Developing sensor-based automation system to measure fabric length in powerloom operations with high precision
- ▶ Designing embedded solution using rotary encoders and microcontrollers to reduce manual measurement errors by up to 95%
- ▶ Implementing real-time display and data logging for production monitoring and inventory management

TECHNICAL SKILLS

Embedded Systems: ESP32, Arduino, LoRa Modules, Sensor Integration, UART/SPI/I2C Protocols

RF & Communication: LoRa Protocol, RF Circuit Design, Antenna Basics, Wireless Communication Systems

Hardware Design: PCB Basics, Circuit Prototyping, Signal Conditioning, Power Management

Software & Tools: Embedded C, Linux (Ubuntu), Git/GitHub, Basic Networking, System Troubleshooting

Domains: IoT Systems, Automation, Disaster Management Tech, Industrial Electronics

ACHIEVEMENTS & LEADERSHIP

- ▶ 🏆 1st Prize – NHIDE 2026 National Hardware & Innovation Design Expo
- ▶ 👥 Team Lead
- ▶ 🎯 Active Member – IEEE Student Branch, BIT-IECC, and AViNYA Innovation Club
- ▶ 💡 Hackathon Participant – Multiple innovation challenges demonstrating rapid prototyping and teamwork

AREAS OF INTEREST

Embedded Systems Design • RF & Wireless Communication • IoT & Automation • Defense Technology • Innovation & Startups